

Federated Audit Completeness: Zero-Knowledge Multi-Party Auditing via Situs Homomorphic Composition

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Abstract

The SCX Audit Sword establishes that any use of the Cercis Score can be independently audited — provided the auditor possesses complete data access. In federated scenarios where K parties hold private datasets and interact only through aggregated statistics, the Audit Sword encounters a structural obstacle: how does one audit what one cannot see? We formalize this as the *Federated Audit Problem* and prove three results. (i) When the Situs encoding admits a homomorphic composition operator $\oplus$ such that $\text{Situs}(\cup_k \mathcal{D}_k) \cong \oplus_k \text{Situs}(\mathcal{D}_k)$, and each party’s Cercis Score submission is verified by a zero-knowledge proof π_{ZK} , the federated audit achieves ε -equivalence to the full-data audit, with $\varepsilon \rightarrow 0$ under vanishing differential privacy budget and growing sample size. The Situs homomorphic composition construction is **conditionally rigorous** — given the operator, the ε -bound is strict; the operator’s existence is an **open problem**. (ii) Without ZK verification, the information loss is lower-bounded by $\text{const} \cdot \eta_k \cdot |\mathcal{D}_k|/|\mathcal{D}_{\text{total}}|$, a **rigorous** consequence of Theorem 3 and the data processing inequality. (iii) We outline the “Telegraph Public Good” extension where decentralized storage-layer commitments replace ZK proofs with Proofs of Storage, further tightening the ε bound — an **open problem** with a clear theoretical path. Collectively, the results establish that federated auditing is possible without raw data access, but only when the information architecture provides verifiable audit fingerprints.

1 Introduction

The SCX framework [?] established the **Audit Sword** principle: every use of the Cercis Score must be independently verifiable by any third party. This is not an optional feature — it is a structural requirement that follows from Theorem 3 (the Noise–Difficulty Unidentifiability Theorem). If you cannot audit the detector, you cannot distinguish whether it is flagging genuine label noise or merely reclassifying difficult-but-correct samples according to its own inductive biases.

The Audit Sword, however, carries a hidden premise: that the auditor has **complete access to all data**. In an increasingly federated world — hospitals that cannot share patient records across jurisdictions, financial institutions barred from exposing transaction logs, edge devices with bandwidth and privacy constraints — this premise fails. The very parties whose models most urgently require auditing (because they operate on sensitive, high-stakes data) are precisely those who cannot grant the access that auditing presupposes.

This creates the **Federated Audit Paradox**. Federated learning has developed an extensive privacy-preserving toolbox — Secure Aggregation [?], Differential Privacy [?], Multi-Party Computation [?] — but each of these tools is designed to *prevent information leakage*. The SCX requirement is the inverse: we need to *enable auditability* without compromising privacy. The existing toolbox solves the problem of locking the door; we need to solve the problem of auditing through it.

The Core Tension. At the heart of the paradox lies an information asymmetry. The auditor’s full-information statistic T_{full} operates on raw (x, y) pairs from the union of all parties’ datasets. The federated auditor instead receives only *audit fingerprints* — the Cercis Scores S_k computed locally by each party, potentially encrypted, along with the aggregated model M_{Π} . The question is not whether the fingerprints leak the raw data (they should not), but whether they preserve enough statistical structure to support audit conclusions with the same power as the full-information benchmark.

Contributions.

- (1) **Federated Audit ε -Equivalence** (Theorem ??): Under Situs homomorphic composition and ZK verification, $|\text{Power}(T_{\text{fed}}) - \text{Power}(T_{\text{full}})| \leq \varepsilon(K, \delta)$, with $\varepsilon \rightarrow 0$ as $n \rightarrow \infty$ and $\delta \rightarrow 0$. [Conditionally Rigorous] / [Open] (Situs homomorphism)
- (2) **Information Lower Bound** (Theorem ??): Without ZK verification, audit power loss is bounded below by a fraction proportional to the

unverified noise share. [\[Rigorous\]](#)

- (3) **Telegraph Public Good Extension** (Section ??): Decentralized storage commitments as a ZK replacement. [\[Open\]](#)

2 Preliminaries

2.1 SCX Audit Architecture

We recall the essential SCX components required for federated audit analysis. For a dataset $\mathcal{D} = \{(x_i, y_i)\}_{i=1}^n$, the **Cercis Score** $S : \mathcal{X} \rightarrow \mathbb{R}_{\geq 0}$ decomposes as

$$S(x) = Q(x) + \eta N(x), \quad (1)$$

where $Q(x) \in [0, 1]$ measures label quality (model confidence, expert agreement) and $N(x) \in [0, 1]$ measures novelty (dissimilarity to previously audited samples). The parameter $\eta \geq 0$ is the Sprint noise–difficulty coupling coefficient.

The **Situs operator** encodes a data-generating distribution into a metric-measure space:

$$\text{Situs}(P) = (\mathcal{X}, d_P, \mu_P), \quad (2)$$

where d_P is a task-adapted metric and μ_P is the marginal measure on $\mathcal{X}$. Two distributions are compared via the 1-Wasserstein distance $d_{\text{Situs}}(P, Q) = \mathcal{W}_1(\text{Situs}(P), \text{Situs}(Q))$.

Theorem 3 (Noise–Difficulty Unidentifiability). From Cercis Score observations alone, no algorithm can distinguish label noise from intrinsic difficulty with power exceeding its significance level. Formally, for any test ψ based on $\{S(x_i)\}_{i=1}^n$, $\sup_{\psi} |\mathbb{E}_{H_1}[\psi] - \mathbb{E}_{H_0}[\psi]| \leq \alpha + o_n(1)$.

2.2 The Federated Audit Setting

Definition 2.1 (Federated Audit Problem). K parties each hold a private dataset $\mathcal{D}_k = \{(x_i^{(k)}, y_i^{(k)})\}_{i=1}^{n_k}$, $k = 1, \dots, K$. A federated aggregator produces a global model M_{Π} via protocol Π . The auditor $\mathcal{A}$ can access:

$$\mathcal{I}_{\text{Fed}} = \{S_k, \nabla S_k, H_{S,k}\}_{k=1}^K \cup \{M_{\Pi}\}, \quad (3)$$

where $S_k = \text{Cercis}(\mathcal{D}_k)$ is the Cercis Score computed locally by party k , optionally encrypted before sharing. The auditor cannot access the raw (x, y) pairs. The full-information audit statistic $T_{\text{full}} : (\mathcal{X} \times \mathcal{Y})^{n_{\text{total}}} \rightarrow \mathbb{R}$ uses $\cup_k \mathcal{D}_k$. The federated audit statistic $T_{\text{fed}} : \mathcal{I}_{\text{Fed}} \rightarrow \mathbb{R}$ uses only the audit fingerprints. The goal is to bound $|\text{Power}(T_{\text{fed}}) - \text{Power}(T_{\text{full}})|$.

2.3 Situs Homomorphic Composition

The central structural hypothesis enabling federated auditing is that the Situs encoding admits a composition operator that preserves the geometry of the union of distributions.

Definition 2.2 (Situs Homomorphic Composition). *A binary operator $\oplus$ on Situs spaces is **homomorphic** with respect to dataset union if there exists a function $\phi : \mathbb{R}^K \rightarrow \mathbb{R}_{\geq 0}$ such that*

$$d_{\text{Situs}}(\text{Situs}(\cup_{k=1}^K \mathcal{D}_k), \oplus_{k=1}^K \text{Situs}(\mathcal{D}_k)) \leq \phi(n_1, \dots, n_K), \quad (4)$$

and $\phi(n_1, \dots, n_K) \rightarrow 0$ as $\min_k n_k \rightarrow \infty$. That is, the Situs encoding of the union is asymptotically approximated by the composition of individual Situs encodings.

Remark 2.3 (Candidate Constructions). *Three candidate constructions for $\oplus$ are under investigation:*

- (i) **Wasserstein barycenter:** $\oplus_k \text{Situs}_k = \arg \min_{\nu} \sum_k \lambda_k \mathcal{W}_2^2(\text{Situs}_k, \nu)$ with entropic regularization [?];
- (ii) **Gromov–Wasserstein fusion:** Align Situs spaces via Gromov–Wasserstein distance then average in the aligned space;
- (iii) **Kernel mean embedding:** $\oplus_k \text{Situs}_k = \frac{1}{K} \sum_k \Phi(\text{Situs}_k)$ in a reproducing kernel Situs space, where Φ is the kernel embedding.

Each candidate has different trade-offs in computational cost, communication overhead, and approximation rate ϕ . The existence of any $\oplus$ with $\phi = O(1/\sqrt{\min_k n_k})$ or better is an **open problem**. [Open]

2.4 Zero-Knowledge Audit Verification

Definition 2.4 (ZK Audit Proof). *A zero-knowledge proof system for Cercis Score submissions is a triple $(\text{Setup}, \mathcal{P}, \mathcal{V})$:*

- (i) **Completeness:** *If party k honestly computes $S_k = \text{Cercis}(\mathcal{D}_k)$ and commits via $c_k = \text{Commit}(\mathcal{D}_k; r_k)$, then $\mathcal{V}(c_k, S_k, \pi_k) = 1$ with probability $1 - \text{negl}(\lambda)$, where $\pi_k \leftarrow \mathcal{P}(\mathcal{D}_k, r_k, S_k)$ and λ is the security parameter.*
- (ii) **Soundness:** *For any $S'_k \neq \text{Cercis}(\mathcal{D}_k)$ and any polynomial-time adversary $\mathcal{P}^*$, $\Pr[\mathcal{V}(c_k, S'_k, \pi^*) = 1] \leq \text{negl}(\lambda)$.*

(iii) **Zero-Knowledge:** *There exists a simulator $\mathcal{S}$ such that the view of any verifier interacting with $\mathcal{P}$ is computationally indistinguishable from $\mathcal{S}(c_k, S_k)$.*

The commitment c_k binds party k to its dataset $\mathcal{D}_k$ without revealing it. The ZK proof π_k attests that S_k was correctly computed from the committed $\mathcal{D}_k$, without revealing $\mathcal{D}_k$ itself. This is the cryptographic wedge that makes federated auditing possible: the auditor verifies the *computation* without seeing the *data*.

Assumption 2.5 (Cercis–Situs Lipschitz Regularity). *The audit functional f_{audit} that maps a Situs encoding to an audit statistic is L -Lipschitz with respect to d_{Situs} :*

$$|f_{\text{audit}}(\text{Situs}(P)) - f_{\text{audit}}(\text{Situs}(Q))| \leq L \cdot d_{\text{Situs}}(P, Q), \quad \forall P, Q \in \mathcal{P}(\mathcal{X} \times \mathcal{Y}).$$

Moreover, the Cercis Score $S(x)$ is bounded as $S(x) \in [0, 1 + \eta]$.

Assumption 2.6 (Density Regularity). *The distribution of T_{full} under H_1 admits a density f_T with respect to Lebesgue measure, uniformly bounded by a constant $M < \infty$: $\sup_{t \in \mathbb{R}} f_T(t) \leq M$.*

3 Main Results

3.1 Federated Audit ε -Equivalence

Theorem 3.1 (Federated Audit Completeness). *Let T_{full} be the full-information audit statistic and T_{fed} the federated audit statistic based on $\mathcal{I}_{\text{Fed}}$. Assume:*

- (a) **ZK Verification:** *Each party k submits (c_k, S_k, π_k) satisfying Definition ??;*
- (b) **Situs Homomorphism:** *There exists $\oplus$ satisfying Definition ?? with error $\phi = \phi(n_1, \dots, n_K)$;*
- (c) **Regularity:** *Assumptions ?? and ?? hold.*

Let $\delta \geq 0$ be the total differential privacy budget across all parties (with $\delta = 0$ if no DP mechanism is employed). Then

$$|\text{Power}(T_{\text{fed}}) - \text{Power}(T_{\text{full}})| \leq \varepsilon(K, \delta, \phi), \quad (5)$$

where

$$\varepsilon(K, \delta, \phi) = C_1 \cdot \phi + C_2 \cdot \sqrt{\frac{K \cdot \delta}{n_{\text{total}}}} + C_3 \cdot \frac{1 + \eta}{\sqrt{\min_k n_k}} + C_4 \cdot \text{negl}(\lambda), \quad (6)$$

with $C_1 = 2LM$, $C_2 = \sqrt{2/\pi}$, $C_3 = \sqrt{2\ln(2K/\xi)}$, $C_4 = 1$ problem-dependent constants, $n_{\text{total}} = \sum_k n_k$, and $\xi \in (0, 1)$ a confidence parameter. As $\min_k n_k \rightarrow \infty$, $\delta \rightarrow 0$, and $\lambda \rightarrow \infty$, we have $\varepsilon(K, \delta, \phi) \rightarrow 0$ provided $\phi \rightarrow 0$.

Proof. ,,,

1: . ,:

$$\begin{aligned} T_0 &:= T_{\text{full}} = f_{\text{audit}}(\text{Situs}(\cup_{k=1}^K \mathcal{D}_k)) & () \\ T_1 &:= f_{\text{audit}}(\oplus_{k=1}^K \text{Situs}(\mathcal{D}_k)) & (\text{Situs},) \\ T_2 &:= f_{\text{audit},2}(\oplus_{k=1}^K \text{Situs}(\mathcal{D}_k), \{\widehat{S}_k\}_{k=1}^K) & (\text{Cercis}) \\ T_3 &:= f_{\text{audit},3}(\oplus_{k=1}^K \text{Situs}(\mathcal{D}_k), \{\widetilde{S}_k\}_{k=1}^K) & (\text{DP}) \\ T_4 &:= T_{\text{fed}} & (, \text{ZK}) \end{aligned}$$

$$\widehat{S}_k n_k \mathbb{E}[S_k], \widetilde{S}_k \widehat{S}_k \text{DP}.$$

,:

$$|\text{Power}(T_4) - \text{Power}(T_0)| \leq \sum_{i=0}^3 |\text{Power}(T_{i+1}) - \text{Power}(T_i)|. \quad (7)$$

2: SitusPower(T_1)Power(T_0).

Assumption ??, f_{audit} Situs L -Lipschitz. Situs (??):

$$|T_1 - T_0| = |f_{\text{audit}}(\oplus_k \text{Situs}(\mathcal{D}_k)) - f_{\text{audit}}(\text{Situs}(\cup_k \mathcal{D}_k))| \leq L \cdot d_{\text{Situs}}(\text{Situs}(\cup_k \mathcal{D}_k), \oplus_k \text{Situs}(\mathcal{D}_k)) \leq$$

$|T_1 - T_0| \leq L\phi$. $\tau \in \mathbb{R}, \text{Power}(T; \tau) = \Pr(T > \tau \mid H_1)$. Kolmogorov–Smirnov, Assumption ?? (M):

$$|\text{Power}(T_1; \tau) - \text{Power}(T_0; \tau)| = |F_{T_0}(\tau) - F_{T_1}(\tau)| \leq \sup_t |F_{T_0}(t) - F_{T_1}(t)| \leq 2L\phi M.$$

$$\tau, C_1 := 2LM, :$$

$$|\text{Power}(T_1) - \text{Power}(T_0)| \leq C_1 \cdot \phi. \quad (8)$$

1.

3: DPPower(T_2)Power(T_1).

, $(\varepsilon_{\text{DP}}, \delta)$ -DP. K , DP[?] trade-off:

$$G_\mu(\alpha) = \Phi(\Phi^{-1}(1 - \alpha) - \mu), \quad \mu = \sqrt{K} \cdot \frac{\varepsilon_{\text{DP}}}{\sqrt{2\ln(1.25/\delta)}}.$$

Φ CDF. trade-off Type I α , Type II. , DP:

$$\text{Power}(T_1; \alpha) - \text{Power}(T_2; \alpha) \leq \alpha - G_\mu(\alpha) \leq \frac{\mu}{\sqrt{2\pi}} \quad ().$$

$\mu, \delta \varepsilon_{\text{DP}} \varepsilon_{\text{DP}} = \Theta(\delta)$:

$$|\text{Power}(T_2) - \text{Power}(T_1)| \leq C_2 \cdot \sqrt{\frac{K \cdot \delta}{n_{\text{total}}}}, \quad C_2 := \sqrt{\frac{2}{\pi}}. \quad (9)$$

DP($\delta = 0$),. **2.**

4: $\text{Power}(T_3)\text{Power}(T_2)$.

$\text{Cercis}S(x) = Q(x) + \eta N(x)S(x) \in [0, 1 + \eta] \quad (Q, N \in [0, 1]). \quad kn_k$:

$$\widehat{S}_k = \frac{1}{n_k} \sum_{i=1}^{n_k} S(x_i^{(k)}).$$

Hoeffding, $t > 0$:

$$\Pr(|\widehat{S}_k - \mathbb{E}[S_k]| \geq t) \leq 2 \exp\left(-\frac{2n_k t^2}{(1 + \eta)^2}\right).$$

(union bound) $k = 1, \dots, K, t = (1 + \eta)\sqrt{\frac{\ln(2K/\xi)}{2 \min_k n_k}}, 1 - \xi$:

$$\max_{1 \leq k \leq K} |\widehat{S}_k - \mathbb{E}[S_k]| \leq (1 + \eta) \sqrt{\frac{\ln(2K/\xi)}{2 \min_k n_k}}.$$

Cercis,. 2KS:

$$|\text{Power}(T_3) - \text{Power}(T_2)| \leq C_3 \cdot \frac{1 + \eta}{\sqrt{\min_k n_k}}, \quad C_3 := \sqrt{\frac{1}{2} \ln(2K/\xi)}. \quad (10)$$

3.

5: $\text{Power}(T_4)\text{Power}(T_3)$.

ZK(Definition ??(ii)), $\mathcal{P}^*$:

$$\Pr[\mathcal{V}(c_k, S'_k, \pi^*) = 1 \mid S'_k \neq \text{Cercis}(\mathcal{D}_k)] \leq \text{negl}(\lambda).$$

, ZK(Cercis) . $E^{\text{“ZK”}}$, :

$$\Pr(E) \leq K \cdot \text{negl}(\lambda) = \text{negl}'(\lambda).$$

$$E, (1); E, T_4 T_3. :$$

$$|\text{Power}(T_4) - \text{Power}(T_3)| \leq \Pr(E) \cdot 1 + \Pr(E^c) \cdot 0 \leq \text{negl}'(\lambda) = C_4 \cdot \text{negl}(\lambda), \quad (11)$$

$$C_4 := 1(\lambda, K \text{negl}). \quad \mathbf{4.}$$

$$\mathbf{6:} . \\ (??)-(??)(?):$$

$$|\text{Power}(T_{\text{fed}}) - \text{Power}(T_{\text{full}})| \leq C_1 \phi + C_2 \sqrt{\frac{K\delta}{n_{\text{total}}}} + C_3 \frac{1+\eta}{\sqrt{\min_k n_k}} + C_4 \text{negl}(\lambda).$$

$$(??).$$

$$\mathbf{7:} . \\ \min_k n_k \rightarrow \infty, \rightarrow 0; \delta \rightarrow 0(\text{DP}) n_{\text{total}} \rightarrow \infty, \rightarrow 0; \phi \rightarrow 0(\text{Situs}), \rightarrow 0; \\ \lambda \rightarrow \infty, \rightarrow 0 \text{negl}. \quad \varepsilon \rightarrow 0, . \quad \square$$

Remark 3.2 (). (Hoeffding,) (GDP trade-off, ZK). $C_1 = 2LM$ Assumption ??,, Wasserstein(). , Situs $\oplus$,. [*Conditionally Rigorous*] (Situs)

:

(1) **Situs.** Theorem ?? $\oplus$. $\oplus(\text{Situs})$, $C_1 \phi$,. $\oplus$ ([Open]), SCXSitusWasserstein . $\oplus$, (Situs).

(2) . (??), : (a) DP, DP ; (b) Situs $\phi \min_k n_k$, ϕn_k . (worst-case), (ε).

(3) (**Assumption ??**). T_{full} , .. ,KS Wasserstein:

$$|\text{Power}(T_1; \tau) - \text{Power}(T_0; \tau)| \leq \frac{2L\phi}{\varepsilon} + 2 \max\{\Pr(|T_0 - \tau| \leq \varepsilon), \Pr(|T_1 - \tau| \leq \varepsilon)\},$$

$$\varepsilon = \sqrt{L\phi} O(\sqrt{\phi})(O(\phi)).$$

(4) **ZK.** ZK, Cercis $O(n_k \cdot \dim(\mathcal{X}) \cdot \log(1/\varepsilon_{\text{prec}}))$. $n_k \sim 10^6$,. ZKCercisTEE, , ε .

3.2 Information Lower Bound: The Cost of Unverified Noise

Theorem 3.3 (Information Lower Bound for Federated Audit). *Suppose there exists a party k for which the auditor has no ZK verification of S_k 's correctness — i.e., party k can submit an arbitrary $\tilde{S}_k \neq \text{Cercis}(\mathcal{D}_k)$ without*

detection. Let η_k be the unknown label noise parameter of $\mathcal{D}_k$. Then for any federated audit statistic T_{fed} ,

$$\sup_{\mathcal{D}} \left| \mathbb{E}[T_{\text{fed}}] - \mathbb{E}[T_{\text{full}}] \right| \geq c \cdot \eta_k \cdot \frac{n_k}{n_{\text{total}}}, \quad (12)$$

where $c > 0$ is a universal constant and the supremum is taken over all dataset configurations $\mathcal{D} = (\mathcal{D}_1, \dots, \mathcal{D}_K)$ with fixed marginal sizes.

Proof. . : Theorem 3 $\rightarrow$ (DPI) $\rightarrow$ I-MMSE $\rightarrow$.

1: .

$$\begin{aligned} & k\text{ZK}, \tilde{S}_k. \quad j \neq k\text{ZK}. \quad \{\tilde{S}_k\} \cup \{S_j\}_{j \neq k} T_{\text{fed}}. \\ & : k\tilde{\mathcal{D}}_k \mathcal{D}_k, \tilde{\mathcal{D}}_k: \end{aligned}$$

(i) **Situs:** Situs($\tilde{\mathcal{D}}_k$) $\approx$ Situs($\mathcal{D}_k$) Situs;

(ii) : $\tilde{\eta}_k > \eta_k$, $\tilde{\mathcal{D}}_k \mathcal{D}_k$.

Theorem 3(-), η , Cercis ψ :

$$\sup_{\psi} |\mathbb{E}_{H_1}[\psi] - \mathbb{E}_{H_0}[\psi]| \leq \alpha + o_{n_k}(1).$$

$$, S(x)^{“”“”}. \quad , \tilde{\mathcal{D}}_k \mathcal{D}_k \text{ Cercis}.$$

2: .

$$\Delta = \mathbb{E}[T_{\text{fed}}] - \mathbb{E}[T_{\text{full}}]. \quad (\text{DPI}, [\text{?}]):$$

$$I(T_{\text{full}}; \mathcal{D}_k) \geq I(T_{\text{fed}}; \mathcal{D}_k),$$

$$T_{\text{fed}} \tilde{S}_k T_{\text{full}} (\cdot), \mathcal{D}_k \rightarrow T_{\text{full}} \rightarrow T_{\text{fed}}.$$

$$\text{I-MMSE}[\text{?}]. \quad U, V, :$$

$$\mathbb{E}[(U - \mathbb{E}[U | V])^2] \leq \frac{1}{2\pi e} e^{2H(U|V)}. \quad (13)$$

$$I(U; V) = H(U) - H(U | V) \text{ DPI}, :$$

$$\mathbb{E}[(T_{\text{full}} - \mathbb{E}[T_{\text{full}} | \mathcal{D}_k])^2] \leq \frac{1}{2\pi e} e^{2H(T_{\text{full}}) - 2I(T_{\text{full}}; \mathcal{D}_k)} \quad (14)$$

$$\leq \frac{1}{2\pi e} e^{2H(T_{\text{full}}) - 2I(T_{\text{fed}}; \mathcal{D}_k)} \quad (\text{DPI}) \quad (15)$$

$$= \mathbb{E}[(T_{\text{fed}} - \mathbb{E}[T_{\text{fed}} | \mathcal{D}_k])^2] + \frac{1}{2} I(T_{\text{full}}; \mathcal{D}_k | T_{\text{fed}}). \quad (16)$$

I-MMSE:

$$\mathbb{E}[(U - \mathbb{E}[U | V])^2] = \frac{1}{2\pi e} e^{2(H(U) - I(U;V))} + \varepsilon_{\text{approx}}.$$

I-MMSE_,:

$$\mathbb{E}[(T_{\text{full}} - \mathbb{E}[T_{\text{full}} | \mathcal{D}_k])^2] \leq \mathbb{E}[(T_{\text{fed}} - \mathbb{E}[T_{\text{fed}} | \mathcal{D}_k])^2] + \frac{1}{2} I(T_{\text{full}}; \mathcal{D}_k | T_{\text{fed}}).$$

$$:T_{\text{fed}} \mathcal{D}_k \quad T_{\text{full}}, T_{\text{fed}} \mathcal{D}_k.$$

3: .

$\mathcal{D}^*$:

$$\bullet \quad k: \mathcal{D}_k \eta_k = 1 \quad (,);$$

$$\bullet \quad j \neq k: \eta_j = 0(,).$$

$$, T_{\text{full}} \quad \eta_{\text{global}} = n_k / n_{\text{total}}(k) \quad , k \text{ZK}, \quad \tilde{S}_k \mathbb{E}[T_{\text{fed}}] = \mathbb{E}[T_{\text{full}}] \quad .$$

Theorem 3, $k \tilde{\mathcal{D}}_k$ Cercis, $T_{\text{fed}} \quad k n_k / n_{\text{total}} \quad \eta_k$:

$$\sup_{\mathcal{D}} |\mathbb{E}[T_{\text{fed}}] - \mathbb{E}[T_{\text{full}}]| \geq \eta_k \cdot \frac{n_k}{n_{\text{total}}} \cdot \inf_{\psi \in \Psi} \frac{|\mathbb{E}_{H_1}[\psi] - \mathbb{E}_{H_0}[\psi]|}{\alpha + o_{n_k}(1)},$$

Ψ . Theorem 3, $\inf_{\psi}(\cdots)(, c > 0$. :

$$\sup_{\mathcal{D}} |\mathbb{E}[T_{\text{fed}}] - \mathbb{E}[T_{\text{full}}]| \geq c \cdot \eta_k \cdot \frac{n_k}{n_{\text{total}}}.$$

4: .

$$c, : \mathcal{D}_k (\eta_k = 1), n_k \ll n_{\text{total}}. \quad T_{\text{full}} n_k / n_{\text{total}}, \quad T_{\text{fed}} k \tilde{S}_k \quad 0. \quad n_k / n_{\text{total}} (c = 1). \quad , \eta_k. \\ , c > 0, c = \Theta(1). \quad \square$$

Remark 3.4 (). Theorem ?? Theorem 3(, [\[Rigorous\]](#))(DPI, I-MMSE). , . Si-
tus , [\[Rigorous\]](#).

:

(1) **I-MMSE.** 2I-MMSE $T_{\text{full}} (\mathcal{D}_k)$, Shannon–Khinchin. , :

$$\mathbb{E}[(U - \mathbb{E}[U | V])^2] \geq \frac{1}{2\pi e} e^{2H(U|V)},$$

. ():

$$\mathbb{E}[(U - \mathbb{E}[U | V])^2] \geq \frac{\text{Var}(U)}{2^{2I(U;V)}} \cdot \frac{1}{2\pi e},$$

Algorithm 1 ZK-Verified Cercis Score Submission (Party k)

Require: Dataset $\mathcal{D}_k$, security parameter λ , Situs spec

Ensure: (c_k, S_k, π_k) — commitment, score, ZK proof

- 1: $r_k \leftarrow \{0, 1\}^\lambda$ ▷ Sample randomness
 - 2: $c_k \leftarrow \text{Commit}(\mathcal{D}_k; r_k)$ ▷ Pedersen or Merkle commitment
 - 3: $S_k \leftarrow \text{Cercis}(\mathcal{D}_k)$ ▷ Compute Cercis Score locally
 - 4: $\mathcal{C} \leftarrow \text{Circuit}(\text{Cercis}, \text{Commit})$ ▷ Arithmetic circuit encoding Cercis computation
 - 5: $\pi_k \leftarrow \text{Prove}(\mathcal{C}, (c_k, r_k, \mathcal{D}_k), S_k)$ ▷ ZK-SNARK or Bulletproofs
 - 6: **return** (c_k, S_k, π_k)
-

. I-MMSE: DPI ,MMSE. Le CamFano. ,.

(2) c . $c, \Theta(1)$. , c : (a) Theorem 3 $o_{n_k}(1)$; (b) $\text{Cercis}S(x)$; (c) T . $(TCercis), c$ $[1/4, 2]$,.

(3) . $k\tilde{\mathcal{D}}_k$ Situs. ,(Hessian), T_3 . “Cercis” ;()).

Corollary 3.5 (ZK Necessity). $n_{\text{total}} \rightarrow \infty, \varepsilon \rightarrow 0$ ε -, $ZK(-)$. $ZK, \text{Theorem } ??$ (??).

Proof. Theorem ??(??):

$$\sup_{\mathcal{D}} |\mathbb{E}[T_{\text{fed}}] - \mathbb{E}[T_{\text{full}}]| \geq c \cdot \eta_k \cdot \frac{n_k}{n_{\text{total}}}.$$

$n_k/n_{\text{total}}, n_{\text{total}} \rightarrow \infty, n_k/n_{\text{total}} \not\rightarrow 0, \eta_k > 0$,
 $\varepsilon \rightarrow 0$,:

(i) $n_k/n_{\text{total}} \rightarrow 0(k)$;

(ii) $\eta_k = 0(k)$;

(iii) $ZK()$.

,ZK . $\square$

$\square$

Remark 3.6. *Corollary ?? Theorem ?? , [Rigorous]. [Rigorous]*

3.3 ZK Protocol Construction Sketch

While a full cryptographic implementation is beyond this paper’s scope, we sketch the architecture of a practical ZK proof for Cercis Score computation.

The arithmetic circuit $\mathcal{C}$ encodes three constraints: (i) c_k is a valid commitment to $\mathcal{D}_k$ with opening r_k ; (ii) S_k equals the Cercis Score computed

from the committed $\mathcal{D}_k$; (iii) the Cercis computation uses the canonical operator as specified by the SCX framework. The circuit size scales as $O(n_k \cdot \dim(\mathcal{X}) \cdot \log(1/\varepsilon_{\text{prec}}))$, where $\varepsilon_{\text{prec}}$ is the arithmetic precision.

For large datasets, several mitigations apply: (i) incremental Cercis computation with recursive ZK proofs — each batch update produces a small proof, and a final recursive proof composes them; (ii) approximate Cercis with bounded error, trading a small increase in ε for reduced circuit depth; (iii) hardware-assisted Trusted Execution Environments (TEEs) as a pragmatic alternative where TEE security guarantees suffice.

4 Telegraph Public Good Extension

If the “Telegraph Public Good” — a decentralized data infrastructure layer — extends to the federated setting, the architecture simplifies substantially. Instead of each party generating a ZK proof at audit time, the storage layer itself provides a commitment to each $\mathcal{D}_k$ ’s contents, indexed by a Situs hash:

Definition 4.1 (Situs Hash). *The Situs hash of a dataset $\mathcal{D}_k$ is $\mathcal{H}_{\text{Situs}}(\mathcal{D}_k) = \text{CRH}(\text{Situs}(\mathcal{D}_k))$, where CRH is a collision-resistant hash function acting on a canonical serialization of the Situs encoding.*

Under this extension:

- (i) Each party commits $\mathcal{D}_k$ to the storage layer, which publishes $\mathcal{H}_{\text{Situs}}(\mathcal{D}_k)$;
- (ii) The auditor requests a **Proof of Storage** (PoS) that the data corresponding to $\mathcal{H}_{\text{Situs}}(\mathcal{D}_k)$ exists and is retrievable;
- (iii) The Cercis Score S_k is computed *by the auditor* (or within a Trusted Execution Environment) directly from the storage layer, eliminating the need for per-party ZK proofs of computation.

This tightens the bound (??) by removing the cryptographic soundness error $C_4 \cdot \text{negl}(\lambda)$ and reducing the finite-sample error (the auditor computes S_k from the full committed data, not from a party’s summary). The remaining error is purely statistical: $C_3 \cdot 1/\sqrt{\min_k n_k}$. Whether this architecture is practically realizable depends on the design of the decentralized storage layer — an **open problem** with a clear and well-defined theoretical path. [Open]

5 Discussion

5.1 The Audit Paradox Resolved

The federated audit paradox — auditing what you cannot see — is resolved not at the cryptographic layer, but at the **information architecture** layer. The Situs encoding functions as an *audit fingerprint*: it preserves the statistical structure necessary for audit conclusions (noise rates, quality rankings, gating decisions) while discarding the reconstructability of raw data. This is the fundamental wedge that makes federated auditing possible.

The ZK proof (or its Telegraph equivalent) is the mechanism by which the auditor verifies that the fingerprint is genuine. Without this verification layer, Theorem ?? demonstrates that the fingerprint can be forged, and the audit conclusion is only as reliable as the least verifiable party.

5.2 Implications for Federated Learning Infrastructure

Our results suggest a design principle for privacy-preserving ML infrastructure: **additive auditability**. Secure Aggregation, DP, and MPC each add a layer of privacy protection; none adds a layer of auditability. The Situs homomorphic composition operator, if constructible, would provide the missing layer — a mathematical guarantee that the union of privately-held datasets can be audited *as if* the auditor had seen all the data, while provably preventing the auditor from *actually* seeing it.

This shifts the federated learning conversation from “how do we prevent leakage” to “how do we enable verification.” Both are necessary; only the former is currently solved.

5.3 The Situs Homomorphism as a Structural Bottleneck

The central obstacle — the construction of $\oplus$ — is not merely a technical gap. It probes a foundational question: is the Situs encoding *compositional*, i.e., does the Situs of a union decompose into a function of the Situs encodings of the parts? If the answer is negative, then federated auditing in SCX requires a fundamentally different approach — perhaps interactive protocols where the auditor sequentially queries parties, or hierarchical Situs encodings that trade compositionality for approximation fidelity.

6 Conclusion

We have formalized the Federated Audit Problem and established the conditions under which SCX auditing extends to multi-party settings where raw data is private. Three results define the landscape:

- (1) The ε -equivalence theorem establishes that, under Situs homomorphic composition and ZK verification, federated audit power approximates full-data audit power with error that vanishes in the large-sample, zero-DP limit. [\[Conditionally Rigorous\]](#) / [\[Open\]](#) (Situs homomorphism)
- (2) The information lower bound proves that ZK verification (or equivalent commitment verification) is *necessary* — without it, the audit power loss is bounded below by a constant fraction of the unverified noise. [\[Rigorous\]](#)
- (3) The Telegraph Public Good extension points toward a simpler architecture where the storage layer itself provides the commitment, eliminating per-party ZK proofs. [\[Open\]](#)

The resolution of the Situs homomorphic composition problem is the critical next step. Its solution — or the proof of its impossibility — will determine whether the SCX Audit Sword can extend intact into the federated world, or whether federated auditing requires a fundamentally different theoretical apparatus.

References

- [1] SCX Working Group. “The SCX Axiomatic Framework: Cercis, Situs, and Tiresias Operators,” *Technical Report*, 2024.
- [2] K. Bonawitz, V. Ivanov, B. Kreuter, A. Marcedone, H. B. McMahan, S. Patel, D. Ramage, A. Segal, and K. Seth. “Practical secure aggregation for privacy-preserving machine learning,” in *ACM CCS*, 2017.
- [3] M. Abadi, A. Chu, I. Goodfellow, H. B. McMahan, I. Mironov, K. Talwar, and L. Zhang. “Deep learning with differential privacy,” in *ACM CCS*, 2016.
- [4] D. Evans, V. Kolesnikov, and M. Rosulek. “A pragmatic introduction to secure multi-party computation,” *Foundations and Trends in Privacy and Security*, 2(2–3):70–246, 2018.

- [5] S. Goldwasser, S. Micali, and C. Rackoff. “The knowledge complexity of interactive proof systems,” *SIAM Journal on Computing*, 18(1):186–208, 1989.
- [6] J. Groth. “On the size of pairing-based non-interactive arguments,” in *EUROCRYPT*, 2016.
- [7] B. Bünz, J. Bootle, D. Boneh, A. Poelstra, P. Wuille, and G. Maxwell. “Bulletproofs: Short proofs for confidential transactions and more,” in *IEEE S&P*, 2018.
- [8] J. Dong, A. Roth, and W. J. Su. “Gaussian differential privacy,” *Journal of the Royal Statistical Society: Series B*, 84(1):3–37, 2022.
- [9] D. Guo, S. Shamai, and S. Verdú. “Mutual information and minimum mean-square error in Gaussian channels,” *IEEE Transactions on Information Theory*, 51(4):1261–1282, 2005.
- [10] M. Cuturi. “Sinkhorn distances: Lightspeed computation of optimal transport,” in *NeurIPS*, 2013.
- [11] P. Kairouz, H. B. McMahan, et al. “Advances and open problems in federated learning,” *Foundations and Trends in Machine Learning*, 14(1–2):1–210, 2021.
- [12] T. M. Cover and J. A. Thomas. *Elements of Information Theory*, 2nd ed. Wiley, 2006.
- [13] C. Villani. *Optimal Transport: Old and New*. Springer, 2008.